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Unit – 1 : Session – 3 : SLO – 2 | Apply Control Structures

■ Activity (Solutions with output)

- 1) Prime check for $n = 7$: loop $i = 2..3$, $7 \% 2 = 1$ and $7 \% 3 = 1$, so flag stays 0 → Output: Prime. Uses selection inside iteration.
- 2) Factorial: $\text{fact} = 1$; for $i = 1..n$, $\text{fact} *= i$ → for $n = 5$: $1 \times 2 \times 3 \times 4 \times 5 = 120$. Uses iteration (sequence of multiplications).
- 3) Largest of a, b, c : if $(a > b \ \&\& \ a > c)$ print a ; else if $(b > c)$ print b ; else print c → for $a=3, b=9, c=5$ → Output: 9. Uses nested selection (if-else ladder).